

ALF basics: Models & Your own model

Example: Dirac fermions coupled to Ising model

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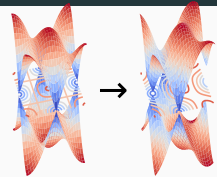
July 15-19, 2024

The Physics: Nematic quantum criticality in Dirac systems

PRL **128**, 157203 (2022) [arXiv:2110.02668]

J. Schwab, L. Janssen, K. Sun, Z. Y. Meng,

I. F. Herbut, M. Vojta and F. F. Assaad



- Nematic phase transition of Dirac Fermions
 - Break rotation symmetry without opening a gap
 - ⇒ Allow meandering of Dirac cones
- Continuous transitions
- No emergent Lorentz symmetry
- Very slow RG-flow
 - ⇒ Numerical (and experimental) results do **not** represent IR fixed point
- Quasiuniversality

The Model

$$\mathcal{H} = \mathcal{H}_0 + \mathcal{H}_{\text{Ising}} + \mathcal{H}_{\text{Yuk}}$$

$\mathcal{H}_0$ Non-interacting Dirac fermions

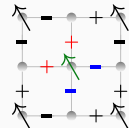
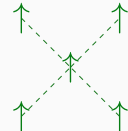
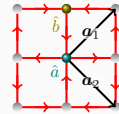
- π -flux square lattice

$\mathcal{H}_{\text{Ising}}$ Transverse-field Ising model

$$\mathcal{H}_{\text{Ising}} = -J \sum_{\langle R, R' \rangle} \hat{S}_R^z \hat{S}_{R'}^z - h \sum_R \hat{S}_R^x$$

$\mathcal{H}_{\text{Yuk}}$ Yukawa interaction

- Couple fermions to spins through terms of form $\alpha \hat{S}_R^z \hat{C}_R^\dagger \hat{C}_{R+\delta}$



$$\hat{\mathcal{H}} = \hat{\mathcal{H}}_T + \hat{\mathcal{H}}_V + \hat{\mathcal{H}}_I + \hat{\mathcal{H}}_{0,I}, \text{ where}$$

$$\hat{\mathcal{H}}_T = \sum_{k=1}^{M_T} \sum_{\sigma=1}^{N_{\text{col}}} \sum_{s=1}^{N_{\text{fl}}} \sum_{x,y}^{N_{\text{dim}}} \hat{c}_{x\sigma s}^\dagger T_{xy}^{(ks)} \hat{c}_{y\sigma s} \equiv \sum_{k=1}^{M_T} \hat{T}^{(k)},$$

$$\hat{\mathcal{H}}_V = \sum_{k=1}^{M_V} U_k \left\{ \sum_{\sigma=1}^{N_{\text{col}}} \sum_{s=1}^{N_{\text{fl}}} \left[\left(\sum_{x,y}^{N_{\text{dim}}} \hat{c}_{x\sigma s}^\dagger V_{xy}^{(ks)} \hat{c}_{y\sigma s} \right) + \alpha_{ks} \right] \right\}^2 \equiv \sum_{k=1}^{M_V} U_k \left(\hat{V}^{(k)} \right)^2,$$

$$\hat{\mathcal{H}}_I = \sum_{k=1}^{M_I} \hat{Z}_k \left(\sum_{\sigma=1}^{N_{\text{col}}} \sum_{s=1}^{N_{\text{fl}}} \sum_{x,y}^{N_{\text{dim}}} \hat{c}_{x\sigma s}^\dagger I_{xy}^{(ks)} \hat{c}_{y\sigma s} \right) \equiv \sum_{k=1}^{M_I} \hat{Z}_k \hat{I}^{(k)}.$$

$$\mathcal{H}_0 = \hat{\mathcal{H}}_T$$

$$\mathcal{H}_{\text{Yuk}} = \hat{\mathcal{H}}_I$$

$$\mathcal{H}_{\text{Ising}} = \hat{\mathcal{H}}_{0,I}$$

To Do

Goal: Write a new Hamiltonian called `Nematic_Dirac_demo`

Steps necessary:

1. Create `Hamiltonian_Nematic_Dirac_demo_smod.F90` in `Prog/Hamiltonians/`
 - Define parameters
 - Create lattice
 - Set up hopping
 - Set up interaction
 - `Op_V`
 - `S0`
 - Define observables
2. Add `Nematic_Dirac_demo` to `Prog/Hamiltonians.list`

Copy template

```
cp Hamiltonian_##NAME##_smod.F90 Hamiltonian_Nematic_Dirac_demo_smod.F90
```

Replace all occurrences of ##NAME## with Nematic_Dirac_demo

Add optional procedure S0 to enable Ising dynamics

Define parameters

Parameter specification:

- Starts with `#PARAMETERS START#` `<namelist name>`
- Ends with `#PARAMETERS END#`
- One parameter per line, with default value
- Optional: Description

Test with

```
../parse_ham.py --test_file Hamiltonian_Nematic_Dirac_demo_smod.F90
```

Declare additional Hamiltonian-wide variables

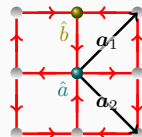
Populate `Ham_set` body

- Read parameters
- Set fixed parameters of Main Hamiltonian
- Setup the Bravais lattice
- Setup the hopping / single-particle part
- Setup the interaction.
- Write parameters to info file

Create Lattice

Bravais lattice:

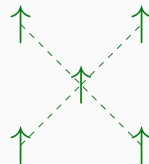
$$\mathbf{a}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \mathbf{a}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$



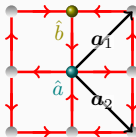
Unit cell:

Fermions: $\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Ising spins: $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$



Hopping

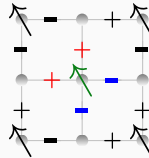


$$\mathcal{H}_0 = -t \sum_R \sum_{\sigma=1}^{N_\sigma} \hat{a}_{R,\sigma}^\dagger \left(\hat{b}_{R,\sigma} e^{-i\frac{\pi}{4}} + \hat{b}_{R+e_-, \sigma} e^{i\frac{\pi}{4}} \right. \\ \left. + \hat{b}_{R+e_- - e_+, \sigma} e^{-i\frac{\pi}{4}} + \hat{b}_{R-e_+, \sigma} e^{i\frac{\pi}{4}} \right) + \text{h.c.},$$

- `call Op_make(Op_T(n_check, nf), size_of_op)`
- Fill operator
- `call Op_set(Op_T(n_check, nf))`

Sum over σ is builtin.

Interaction



$$\mathcal{H}_{\text{Yuk}} = -\xi \sum_R \sum_{\sigma=1}^{N_\sigma} \hat{s}_R^z \hat{a}_{R,\sigma}^\dagger \left(\hat{b}_{R,\sigma} e^{-i\frac{\pi}{4}} - \hat{b}_{R+\mathbf{e}_-, \sigma} e^{i\frac{\pi}{4}} \right. \\ \left. - \hat{b}_{R+\mathbf{e}_- - \mathbf{e}_+, \sigma} e^{-i\frac{\pi}{4}} + \hat{b}_{R-\mathbf{e}_+, \sigma} e^{i\frac{\pi}{4}} \right) + \text{h.c.},$$

- `call Op_make(Op_V(n_int, nf), size_of_op)`
- Fill operator
- `call Op_set( Op_V(n_int, nf) )`

Sum over σ is builtin.

Ising action: Mapping to $d+1$ -dimensional classical Ising model

$$\mathcal{H}_{\text{Ising}} = -J \sum_{\langle R, R' \rangle} \hat{S}_R^Z \hat{S}_{R'}^Z - h \sum_R \hat{S}_R^X,$$

$$Z = \text{Tr} [\exp(-\beta H_{\text{Ising}})]$$

Trotter decomposition:

$$Z = \sum_{\Phi} \left\langle \Phi \left| \left(\exp \left(\Delta_{\tau} J \sum_{\langle R, R' \rangle} \hat{S}_R^Z \hat{S}_{R'}^Z \right) \exp \left(\Delta_{\tau} h \sum_R \hat{S}_R^X \right) \right)^{L_{\tau}} \right| \Phi \right\rangle + \mathcal{O}(\Delta_{\tau}^2)$$

State: $d \rightarrow d + 1$ dimensions

$$Z = \sum_C \prod_{t=1}^{L_{\tau}} \left[\exp \left(\Delta_{\tau} J \sum_{R, R'} s_{R,t}^Z s_{R',t}^Z \right) \left\langle \Phi_t \left| \exp \left(h \Delta_{\tau} \sum_R \hat{S}_R^X \right) \right| \Phi_{t+1} \right\rangle \right]$$

Ising action

$$Z = \lambda^{NL_\tau} \sum_C e^{-S_{\text{Ising}}(C)} + \mathcal{O}(\Delta_\tau^2)$$

$$e^{-S_{\text{Ising}}(C)} = \prod_{t=1}^{L_\tau} \left[\exp \left(\Delta_\tau J \sum_{R,R'} s_{R,t}^z s_{R',t}^z \right) \exp \left(\gamma \sum_R s_{R,t}^z s_{R,t+1}^z \right) \right]$$

$$\gamma = -\frac{1}{2} \ln(\tanh h\Delta_\tau) \quad \lambda = \sqrt{\sinh(h\Delta_\tau) \cosh(h\Delta_\tau)}$$

Single spin flip:

$$\frac{e^{-S_{\text{Ising}}(C_{\text{new}})}}{e^{-S_{\text{Ising}}(C_{\text{old}})}} = \begin{cases} \tanh(\Delta_\tau h) & s_{R,t}^z s_{R,t+1}^z : 1 \rightarrow -1 \\ 1/\tanh(\Delta_\tau h) & s_{R,t}^z s_{R,t+1}^z : -1 \rightarrow 1 \\ \exp(-2\Delta_\tau J) & s_{R,t}^z s_{R',t}^z : 1 \rightarrow -1 \\ \exp(2\Delta_\tau J) & s_{R,t}^z s_{R',t}^z : -1 \rightarrow 1 \end{cases}$$

Create Observables: Alloc_obs

Allocate the storage space for Observables

- Part: Fermion particle number
- ising_z: $\langle \hat{S}^z \rangle$
- Kin_Pot_E: $\mathcal{H}_0, \mathcal{H}_{\text{Yuk}}, \mathcal{H}_0 + \mathcal{H}_{\text{Yuk}}$
- ising_x: $\langle \hat{S}^x \rangle = \langle \partial_h \mathcal{H} \rangle$
- m: $\sum_R \langle \hat{S}_0^z \hat{S}_R^z \rangle$ and its 2nd, 4th moment
- IsingX: $\sum_{R'} \langle \hat{S}_{R'}^x \hat{S}_{R'+R}^x \rangle$
- Green: $\sum_{R'} \langle \hat{c}_{R'}^\dagger \hat{c}_{R'+R} \rangle$
- Den: $\sum_{R'} \langle \hat{n}_{R'} \hat{n}_{R'+R} \rangle$

Calculate equal time observables: Obser

Scalar observables

- `Obs_scal(I)%N`: Number of measurements
- `Obs_scal(I)%Ave_sign`: Average sign
- `Obs_scal(I)%Obs_vec(n)`: Measured value

Correlation functions

- `Obs_scal(I)%N`: Number of measurements
- `Obs_scal(I)%Ave_sign`: Average sign
- `Obs_scal(I)%Obs_Latt0(no)`:
Background $\langle \hat{O}_{R,no} \rangle$
- `Obs_scal(I)%Obs_Latt(imj, nt, no1, no2)`:
Correlation $\langle \hat{O}_{R,no1} \hat{O}_{R+R(imj),no2} \rangle$

Calculate time-displaced observables: `ObserT`

Analogous to equal time correlation functions

Finishing

- Add `Nematic_Dirac_demo` to `Prog/Hamiltonians.list`
- (Try to) compile
- Test run

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Tomorrow: Extract results vi `pyALF`